

Quantum Computation and Information

Lecture 2: Quantum Circuit

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle \mapsto U |\psi\rangle$$

- quantum logic gates
- quantum circuit
- universal quantum gates

quantum gates

- basis of Hilbert space: $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

- basis for linear operators (identity and Pauli X, Y, Z matrices)

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$



- another basis of Hilbert space

$$|+\rangle = H |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = H |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

- quantum ‘interference’

$$\begin{aligned}
 H|+\rangle &= \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) \\
 &= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\
 &= \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle + \frac{1}{2}|0\rangle - \frac{1}{2}|1\rangle \\
 &= |0\rangle
 \end{aligned}$$

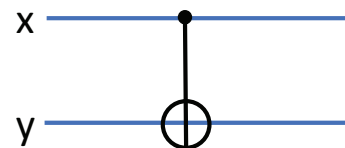
$$\begin{aligned}
 H|-\rangle &= \frac{1}{\sqrt{2}}(H|0\rangle - H|1\rangle) \\
 &= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) - \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\
 &= \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle - \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle \\
 &= |1\rangle
 \end{aligned}$$

– amplitudes can interfere constructively or destructively

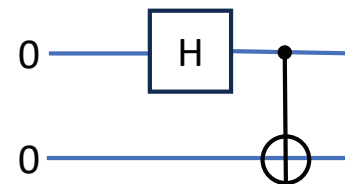
- phase shift gate $P(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$

- controlled-NOT (CNOT) gate mapping $|x, y\rangle$ to $|x, x \oplus y\rangle$

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

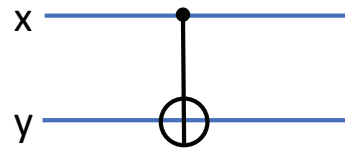


- exercise

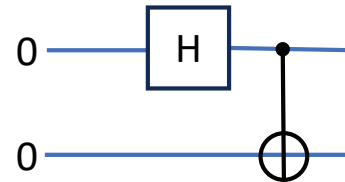


- controlled-NOT (CNOT) gate mapping $|x, y\rangle$ to $|x, x \oplus y\rangle$

$$CCNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



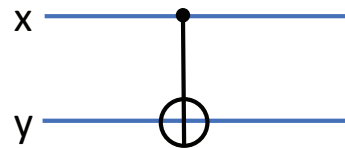
- exercise



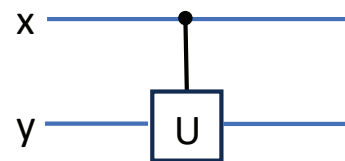
$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

- controlled-NOT (CNOT) gate mapping $|x, y\rangle$ to $|x, x \oplus y\rangle$

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

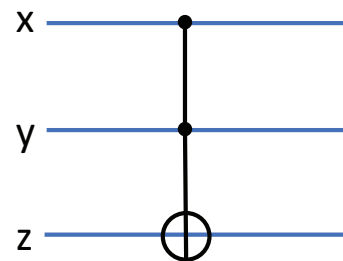


- control-U gate $|0\rangle \langle 0| \otimes I + |1\rangle \langle 1| \otimes U$



- Toffoli gate (CCNOT) mapping $|x, y, z\rangle$ to $|x, y, z \oplus (x \wedge y)\rangle$

$$CCNOT = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$



- a universal reversible gate for classical computing: any classical reversible circuit can be constructed from it

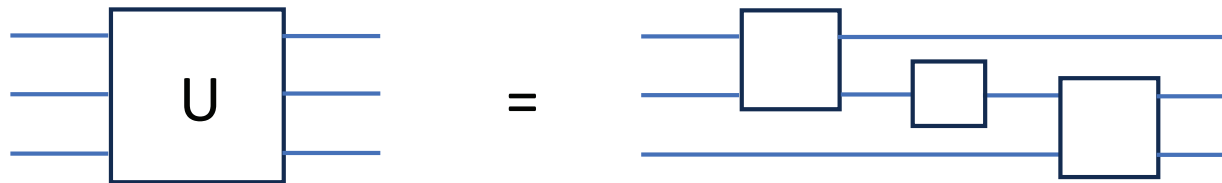
- measurement



quantum circuit

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle \quad \mapsto \quad U |\psi\rangle$$

- a procedure to implement a complicated $2^n \times 2^n$ unitary transformation $U(2^n)$ on n qubits by decomposing it into a product of simple unitary operations on a few qubits at a time



- a generalization of the circuit model of classical computation

quantum circuit

$$|000 \dots 0\rangle \mapsto U |000 \dots 0\rangle$$

- initialization: $|x\rangle = |0\rangle^{\otimes n} = |000 \dots 0\rangle$
 - simple preparation, e.g., lowest energy state of quantum register
- universal set of quantum gates $\{U_1, U_2, \dots, U_m\}$
 - a fine set of gates, each on a small number of qubits, that can be used to approximate any n-qubit unitary to any desired accuracy
- classical control: a classical computer builds a circuit and directs its execution
 - the hybrid quantum-classical paradigm

- read out of qubits in the standard basis $\{|0\rangle, |1\rangle\}$
 - simple measurement
 - complexity of complicated measurement accounted in complexity of an algorithm
 - delayed measurement v.s. measurement at intermediate stages (with subsequent execution conditioned measurement outcomes)
 - * delayed measurement is general
 - * measurement at intermediate stages in practice

universal gates

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle \quad \mapsto \quad U |\psi\rangle$$

- a finite set of quantum gates that can be used to construct a circuit V to approximate any unitary transformation U on any number of qubits to any desired accuracy ϵ

$$\|V - e^{i\theta}U\|_{\infty} \leq \epsilon$$

- exact universality: requires uncountable gate set; e.g., the set of two-qubit gates, any entangling two-qubit gate plus the set of one-qubit gates
- generic universality: almost any two-qubit gate is universal if it can be applied to any pair of qubits
- example choices of universal gate set: $\{CCNOT, H, P(\pi/2)\}$, $\{H, \text{control} - e^{-i\pi/4}P(\pi/2)\}$

Solovay-Kitaev theorem

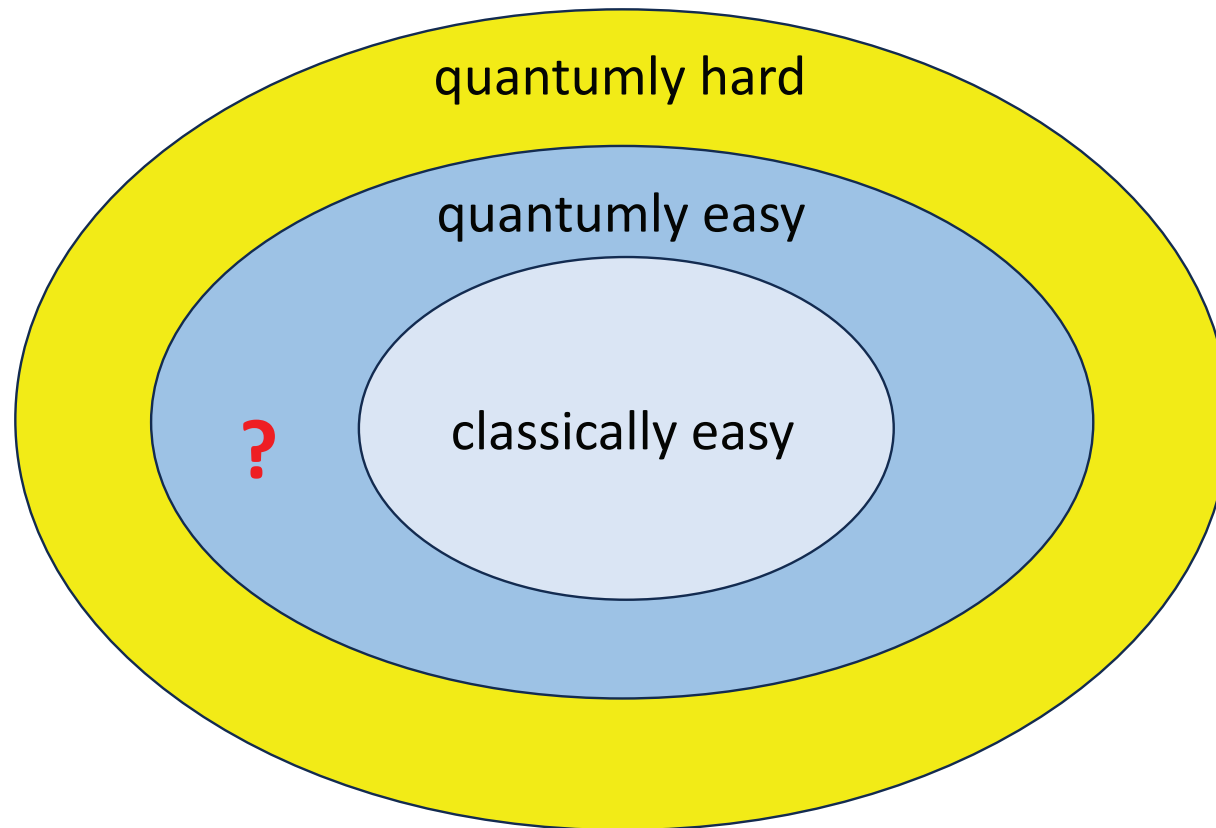
- how large a quantum circuit V is needed to approximate a given unitary transformation U to the desired accuracy ϵ ?
- how large a classical circuit is needed to find such a quantum circuit?
- Solovay-Kitaev theorem: given any universal set of quantum gates, the answer to both questions is $\log^{O(1)} 1/\epsilon$

a fundamental question

- is quantum computing (much) more powerful than classical computing?
 - (reversible) classical computing is a special case of quantum computing
 - quantum computer cannot do anything that classical computer can't do
 - quantum computing “believed” to be more powerful
 - * some problems that are believed to be classically hard are quantumly easy, e.g., factoring
 - * do not know how to simulate a quantum system/computer efficiently using a classical computer
 - * but its power is limited, e.g., may not be able to efficiently solve worst-case instances of NP-hard optimization problems
 - quantum complexity

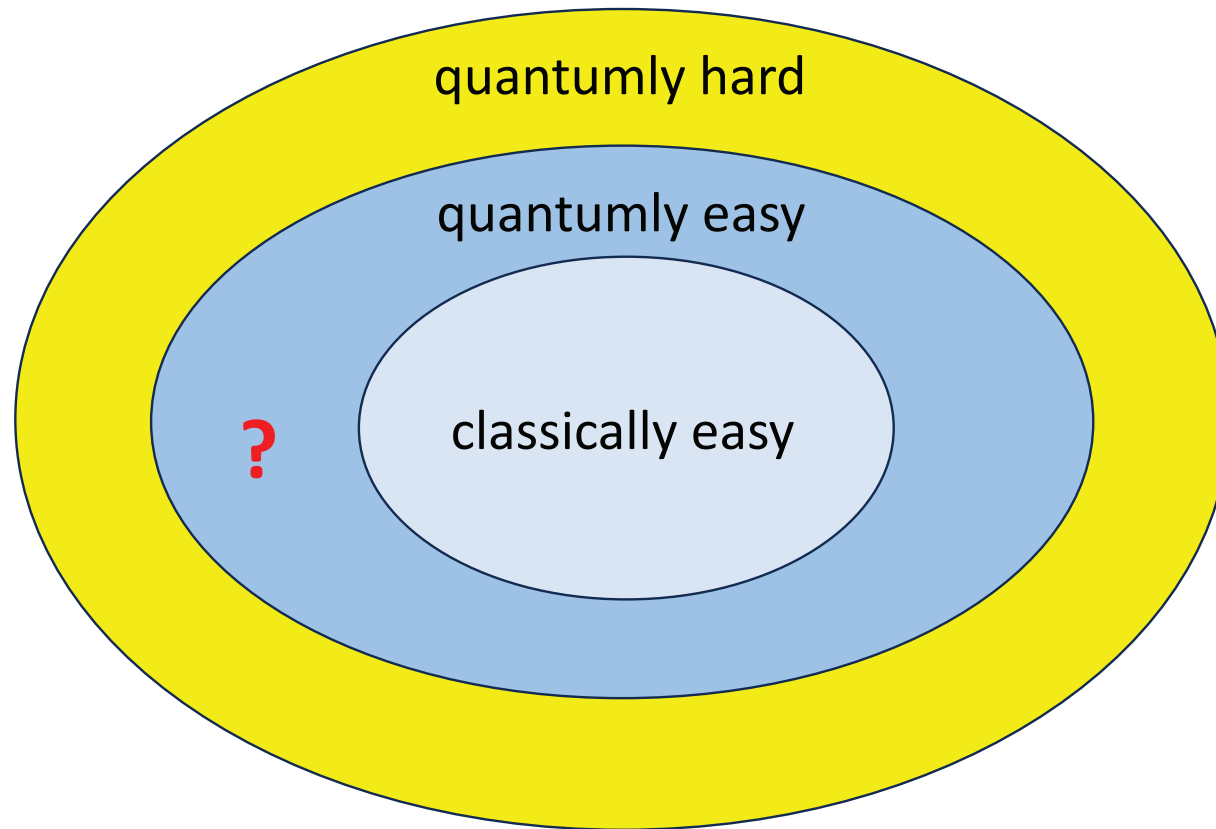
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– factoring, quantum simulation, ...